

CONTINUATION

CONTINUATION — HelixTranslate session-resumption file

Revision: 91 Last modified: 2026-06-17T01:30:00Z

 **RESUME HERE (2026-06-17 ~01:30) —  7 commits pushed (98e0450) · 5 real fixes + §11.4.40 GREEN · LLMsVerifier seed-at-boot WORKING on nezha (count 0→2) · verify-completion being implemented · helixtranslate NON-degraded (never-degrade guard holds: 0 VERIFIED yet)**

SHORT (rev91): HEAD **98e0450** (all 4 remotes synced). Operator authorized "wire it all + do it all". Since rev90, 3 more commits via parallel worktree subagents (each independently reviewed GO, merged main-loop): 2a36a57 (redis ListSessions logs swallowed errors + fallback residual-limitation documented §11.4 observability) → 98e0450 (handlers propagate c.Request.Context()+5min bound — audit HIGH context-misuse, all 6 sites proven SYNC §11.4.102, mutation-proven guard). **7 session commits total** (99bb7a8→a99327b→989f5d1→62cbb28→4223e7c→2a36a57→98e0450). §11.4.40 retest GREEN at HEAD; -race CLEAN. **LLMsVerifier (operator-authorized) — seed-at-boot now WORKING on nezha:** verifier image rebuilt WITH seed code (CGO amd64, rollback img llm-verifier:nezha-prev-20260617T003159Z, src staged /home/milosvasic/helix-system/llmsverifier-src), config.yaml rendered (deepseek+groq, secrets never printed), /api/models count **0→2** (seed log providers +2 models +2). Evidence subagent report. **BUT both models pending, NOT verified** → never-degrade guard HOLDS (helixtranslate gate needs verified → NOT enabled, translation stays on working direct path). **Verify-completion feature being implemented** (verifier VerifyModelHandler only inserts a "running" row, never runs the real ModelVerificationService.VerifyModel nor persists status — subagent wiring it w/ TDD). **NEXT:** review the verify-completion impl → commit to verifier submodule + bump gitlink → rebuild nezha → run real verify pass → INSPECT scores → enable helixtranslate ONLY if a sink-side translation through a verified model proves correct (§11.4.69/§11.4.107 quality gate); else leave NON-degraded + operator.

SHORT (rev90): HEAD **62cbb28** (all 4 remotes synced). Since rev89, added a 4th commit 62cbb28 — **api-server /health + /api info now report version.AppVersion** (were hardcoded "3.0.0"/"1.0.0"; §11.4.108 lie-class) + §11.4.135 deterministic guard `TestHealthCheckReportsAuthoritativeVersion` (RED-on-revert proven). Independent design review (/Volumes/T7/helix-build/qa/design_version_hygiene2.md) corrected the audit: `grpc-server/unified-translator/server` ALREADY report 2.3.1 via `const appVersion=version.AppVersion` (Go `-ldflags -X` can't set a `const` → Makefile `ldflag` is a harmless no-op; Makefile cosmetic `VERSION=3.0.0` left for operator). 4 session commits: 99bb7a8 (version-const+SQLite+scanner) → a99327b (CONTINUATION+verifier-block) → 989f5d1 (SQLite deterministic guard) → 62cbb28 (api-server version). -race sweep CLEAN on 6 concurrency pkgs. **LLMsVerifier wiring still HONEST-BLOCKED (operator)** — see docs/qa/llmsverifier_wire_attempt_20260616T203122Z/STATE.md. NEXT = operator items below.

◆ **ROBUSTNESS FOLLOW-UPS (audit-found this session, NOT fixed — need investigate-before-fix; evidence /Volumes/T7/helix-build/qa/audit_robustness2.md)**

- **HIGH** — **handler request-context misuse:** `pkg/api/handler.go:277/438/575/924` + `batch_handlers.go:145/275` use `context.Background()` (no client-disconnect cancellation, no timeout → a stuck provider hangs the handler). **DO NOT blind-fix:** must FIRST determine per-site whether the translate path is ASYNC (returns `session_id`, translation must OUTLIVE the request → `Background()` is CORRECT) vs SYNC (§11.4.102). If sync, propagate `c.Request.Context()` + bounded timeout (the sites at `handler.go:1166/1251/1429/1609` already do `WithTimeout(5min)`). If async, leave `Background()` but consider a generous standalone timeout. Wrong fix = cancels in-flight book translations on disconnect = regression.
- **MED:** `pkg/distributed/fallback.go:215` abandoned goroutine runs to completion after waiter times out (load amplification); `pkg/storage/redis.go:111/117` `ListSessions` silently continues on errors (anti-bluff observability gap).
- **LOW:** unchecked map type-assertions (`multi_llm.go:173/200/206/207/220`, `handler.go:1564`, `verifier_handlers.go:271/272` — safe-by-construction today, panic-on-drift); `models/downloader.go:109` uncancellable download; library panics (`security/auth.go:31`, `models/registry.go:47/70`, `logger.go:220` `os.Exit`) — return-error candidates.
- **Verified CLEAN:** postgres pool bounded 25/5 + ctx queries; all LLM providers propagate ctx + close bodies; `ssh_pool` ctx-cancellation OK; no stray `sql.Open/http.DefaultClient`.

Untracked test fixtures (§11.4.124 — RESOLVED: LEAVE)

`test/assets/crow_original.md` + `test/fixtures/ebooks/{sample,spanish_sample}_original.md`: NOT referenced by any test (no fresh-clone failure), never tracked, with gitignored `*_translated` output siblings → transient round-trip inputs (possibly another stream's live input). LEFT untouched (not committed, not deleted). Evidence /Volumes/T7/helix-build/qa/fixture_provenance.md.

 (rev89) **RESUME HERE (2026-06-16 ~20:40) —**
 OVERNIGHT STABILITY BATCH
committed+pushed (99bb7a8) · 2 real bugs fixed +
§11.4.40 GREEN · LLMsVerifier wiring HONEST-
BLOCKED (operator) · nezha NON-degraded

SHORT: Read this file + .remember/remember.md FIRST, then `git fetch --all --prune`. HEAD **99bb7a8** (all 4 remotes synced: origin/github/upstream/githubhelixdevelopment). 🍷 `helix_translate-2.3.1` still the release tag (main+10 submodules). **This overnight loop fixed 2 REAL bugs + a scanner false-positive, all reviewed GO + proven, committed 99bb7a8 (FF-pushed):** (1) `pkg/version/app.go` `AppVersion 2.3.0→2.3.1` (was RED — VERSION bumped at 236bac8 without the const); (2) `pkg/storage/sqlite.go` `SQLITE_BUSY` concurrency (unbounded pool + no `busy_timeout` → 110/640 concurrent writes failed; fix `_busy_timeout=5000` + default `SetMaxOpenConns(1)`, RED→GREEN 5x §11.4.85); (3) `credential_scan.sh` HTML-escaped-redaction false-positive + ignore-list its own regression test + new scripts/testing/`test_credential_scan.sh` §11.4.135 guard. **§11.4.40 full retest GREEN at the batch** (sweep 9/9 + build + go test ./... all pass; evidence /Volumes/T7/helix-build/qa/overnight_retest4_*.log). **LLMsVerifier wiring = HONEST-BLOCKED (operator)** — live verifier image (built 15:46Z) PREDATES the seed commit 6e6b0309 (19:26Z); no reproducible on-nezha rebuild context; `config.yaml` is operator-owned + secret-bearing + Permission-denied (can't populate `llms`: without handling secrets blind §11.4.10). NOT enabled, nezha NON-degraded. Operator options in docs/qa/`llmsverifier_wire_attempt_20260616T203122Z/STATE.md`. **NEXT = operator decision on verifier (rebuild-with-seed + populate config + verify-pass) + the version-display-hygiene follow-ups below.**

 ENVIRONMENT (this session, FACT)

- **Root/Data volume was ~100% full** (operator data + a concurrent `helix_code` session) — broke the build harness with `ENOSPC` repeatedly. Worked around by routing ALL Go temp/cache to T7: `export TMPDIR=/Volumes/T7/tmp GOTMPDIR=/Volumes/T7/tmp GOCACHE=/Volumes/T7/helix-build/go-cache XDG_CACHE_HOME=/Volumes/T7/.cache`. Reclaimed Homebrew+SwiftPM caches (regenerable). Did NOT delete operator data / the stopped 41G podman machine (§11.4.122). If `ENOSPC` recurs: `find /private/tmp/claude-501 -path '*/tasks/*.output' -mmin +4 -delete` (NEVER without `-mmin +4` — deletes the in-flight output).

◆ **VERSION-DISPLAY-HYGIENE FOLLOW-UPS (audit-found, NOT fixed — operator-judgment, tracked)**

Same defect class as the version-const bug, but display-only + touch the build-version question CLAUDE.md flags as deliberate ("Makefile references 3.0.0 — treat VERSION as authoritative"):

- Makefile:5,20 VERSION=3.0.0 + -ldflags -X main.appVersion=\$(VERSION) → shipped grpc-server/unified-translator binaries report **3.0.0** not 2.3.1 (§11.4.108 SOURCE→ARTIFACT gap; no test catches the ldflags path).
- cmd/api-server/main.go:582 /health hardcodes "version":"3.0.0"; pkg/api/handler.go:214,223 serve "version":"1.0.0".
- The version gate (app_test.go + bash CM-VERSION-SINGLE-SOURCE) only matches the = form in cmd/ — blind to ldflags, : map form, pkg/. Broaden + add a post-build <bin> -version==VERSION regression guard. (Design notes were to land in /Volumes/T7/helix-build/qa/design_version_hygiene.md but that subagent was rate-limited; re-run or do main-loop.) Evidence: /Volumes/T7/helix-build/qa/audit_hygiene.md.

● **RELEASE + LLMsVERIFIER ARC (this session, post-rev87)**

- 🍌 **helix_translate-2.3.1 RELEASED** — patch fix-cycle tag on main (VERSION 2.3.1) + 10 owned submodules (containers, challenges, doc_processor, llm_orchestrator, llm_provider, vision_engine, llms_verifier, constitution, docs_chain, security), pushed FF to every remote (NO force §11.4.113). EXCLUDED: helix_qa (§11.4.119) + helix_agent (incorporated-not-integrated). Pre-tag §11.4.40 GREEN (sweep 9/9 + build + go test 55 pkgs). Evidence docs/qa/release_helix_translate-2.3.1_20260616T182714Z/. Shipped: nezha 6-svc boot + 6 product defects fixed (commentary-contamination, create-default env-override, verifier 404→503, /health IDLE, unsupported-provider→400); 6-site Translate-arg DATA-LOSS bug-class (4fbc581); multipass+FB2 (c2aa7c8); MINOR-W6-1 (7cabe3f); /providers static-list bug (388a2eb, 19 real providers); secret-hygiene (04f71e9); video 43→66.
- 🚫 **LLMsVerifier populate-then-wire — HONEST-BLOCKED (no bluff, no degrade):**
 - Separate nezha stack llmsverifier (compose dir /home/milosvasic/helix-system/llmsverifier, container llmsverifier_llm-verifier_1, image llm-verifier:nezha). IP drifted .76→.77 but irrelevant — port bound **loopback-only** 127.0.0.1:8080. /api/health 200 healthy; /api/models & /api/providers both **count:0**. Provider keys (GROQ/DEEPSEEK/GOOGLE) ALREADY present in the container env → keys are NOT the blocker.
 - **Root cause (FACT, llm-verifier/api/server.go Start()):** the deployed server mounts EXACTLY 5 hardcoded read routes — GET /api/health, GET /api/models (reads DB, not config), GET /api/models/, POST /api/models/{id}/verify (needs model already in DB), GET /api/providers. NO create/seed route, NO /api/v1/*, NO /auth/login (all → 404, proven). runServer() does NOT seed config.yaml llms: (which is []) into the DB. CLI models create/import POST to /api/v1/models → 404 on this build. batch run is a STUB. The only populate path = direct write to the encrypted SQLite DB (DATABASE_ENCRYPTION_KEY operator-owned, 600-perm .env) →

operator-owned config I must NOT autonomously mutate (§11.4.122/§11.4.10/§9.2).

- **Decision:** DID NOT network connect, DID NOT set LLMSVERIFIER_ENABLED, NO helixtranslate compose/env change. Confirmed safe: GET /api/v1/verified-models → "integration is disabled"; working translation healthy en→sr Добро јутро, како сте? (bridge llm-mistral).

NEXT (priority order)

1. **Operator decision on the verifier image** (§11.4.66) — Option A (recommended): rebuild llm-verifier:nezha so runServer() seeds config.yaml llms: into the DB at boot (or add a seed/verify-all subcommand writing via database.CreateModel, NOT the HTTP client), populate config.yaml llms: (deepseek+groq), rebuild+restart, POST /api/models/{id}/verify → real score → count>0; THEN run the verified helixtranslate wiring (network connect + LLMSVERIFIER_ENABLED=true + LLMSVERIFIER_API_URL=http://llm-verifier:8080 — note verifier must publish on llmsverifier_default, not loopback-only — + scripts/nezha-deploy.sh reboot + sink-side re-validate /api/v1/verified-models real data). Option B: operator seeds the encrypted DB directly.
 2. Known non-blocking (post-2.3.1, recorded not in tag): see .remember/remember.md.
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(prior) RESUME HERE (2026-06-16 ~20:20) — **BUG-CLASS FIXES + VIDEO 54→66 + 4-ITEM** **OPERATOR-REVIEW QUEUE ALL CLEARED** **(live-validated) · NEXT = bug-class review + full** **§11.4.40 retest → release-readiness**

SHORT: Read this file + .remember/remember.md FIRST, then git fetch --all --prune. HEAD d7b4407 (all 4 remotes synced). Live nezha stack rebuilt onto image **08900424e481** (all 6 services healthy incl. monitor — the stranded-monitor bug is fixed). The 4-item operator-review queue is **CLEARED**. NEXT: independent review of the §11.4.69/§11.4.115 bug-class fixes (4fbe581 6-site Translate-arg data-loss + 7cabe3f MINOR-W6-1) → full §11.4.40 retest from last tag → release-readiness (operator tag decision per §11.4.151).

● **THIS SESSION ARC (since the rev86 nezha heavy-testing arc — all §11.4.142-reviewed GO, live-validated, no bluff)**

Live image progression: 2bb4de5 (FB2-fix) → **08900424e481** (this session: / providers fix + all app services recreated). HEAD d7b4407.

• **PART-A/B/C + bug-class fixes (pre-this-commmitter-stream, on HEAD before f6b1f1f):**

- c2aa7c8 — **BUG-MULTIPASS-DEFAULT-MODEL** (resolvePolisherModel substitutes a provider-valid model so -multipass genuinely polishes; honest **✗** on real failure) + **BUG-FB2-HARDCODED-LANG** (FB2 handler honors source_lang/target_lang, validates→400 on unknown). RED→GREEN §11.4.115.
- 04f71e9 — security: scrubbed a leaked SSH password from the tracked tree (§11.4.10/§11.4.30/§11.4.124), closed GAP1/GAP2.
- 4fbc581 — **6-site Translate-arg data-loss bug-class** (empty-payload + wrong-content sent to the LLM across the ebook pipeline) §11.4.69/§11.4.115/§11.4.135.
- 7cabe3f — **MINOR-W6-1** parser chapter-title duplication §11.4.115/§11.4.135.
- af2ef7f + f6b1f1f — §11.4.153 video waves 4–9: **video-confirmed 43→54→66** (+12 net-new waves 7–9, of which 3 genuine PENDING→Confirmed row-flips), §11.4.138 multipass demote, + the cmd/grpc-translate-probe row.

• **THIS BACKGROUND-COMMITTER STREAM (HEAD f6b1f1f → d7b4407), all 4 operator-review items + drift cleared:**

- e5051a5 — **#1 Status video-count drift reconcile → 66** (§11.4.6/§11.4.60/§11.4.91): the headline/Anti-bluff-note/Coverage cell were 66 but 3 caveat cells lagged (Status.md "30 feature rows"; Status_Summary Page-1 "30 of 494", Page-2 "43/496") — all reconciled to 66 with explicit §11.4.6 cell-reconciliation notes; docs_chain features verify=in-sync. (docs_chain CANNOT catch internal contradictions — this was a real release-gate blocker.)
- 0246851 — **#2 FB2 live re-validation evidence** (§11.4.108/§11.4.123/§11.4.83): on nezha 2bb4de5 image, POST /api/v1/translate/fb2 target_lang=es→real Spanish EPUB ("Capítulo de Prueba"...), de→German (distinct), klingon→400. docs/qa/fb2_revalidate_20260616T163552Z/.
- 388a2eb — **#3 /api/v1/providers REAL BUG fixed** (§11.4.115/§11.4.135/§11.4.6): listProviders served a hardcoded static {openai,anthropic,zhipu,deepseek} list ignoring config → now serves the REAL configured set (configured/available/requires_api_key/model + <PROVIDER>_API_KEY env), complete-catalogue fallback when unconfigured. RED→GREEN polarity test + guard HTQ-FIX-006. Independent review NO-GO→GO (§11.4.134). (cmd/api-server getProviders gRPC path untouched.)
- af23440 — **#5b deploy-script dependency-ordering fix** (§11.4.108): no-arg nezha-deploy.sh reboot now stop+rm+recreates ALL app services (grpc/api/server/monitor) onto the fresh image (was gated on a named-service arg → left dependents, esp. monitor, on the stale image). Companion doc Rev 2 + exports.

- 702d907 — **#4 pkg/api/server.go honest §11.4.124 N/A** + record / providers fix in Status (the alt api. Server is NEVER-WIRED test-only scaffolding — kept-not-deleted per §11.4.122/§11.4.124, documented N/A, NOT silently shown as a shipping API). Status Rev 18.
- d7b4407 — **#5a + #3 sink-side re-validation** (§11.4.108/§11.4.69): rebuilt nezha (image 08900424e481) via the fixed reboot → ALL services incl. **monitor** recreated + healthy; GET /api/v1/providers returns **19 config-driven providers** (NOT the old static 4; openai correctly absent — no key), new fields present. docs/qa/providers_revalidate_20260616T170812Z/.

LIVE STATE ANCHORS (§11.4.6, moment-valid)

- **HEAD = d7b4407**, all 4 remotes (origin/github=milos85vasic/Translator.git + HelixDevelopment/HelixTranslate.git, upstream, githubhelixdevelopment) synced. NOTE: origin & github push to the SAME repo pair → push them **SERIALIZED** (parallel pushes race the ref lock; harmless "remote rejected: cannot lock ref" = the loser of the race, re-fetch confirms the tip).
- **Nezha live stack** (nezha.local, ssh host alias nezha.local, repo synced at /home/milosvasic/helixtranslate — a NON-git copy, NOT a clone): image **08900424e481**, all 6 helixtranslate-* containers healthy (server-TLS :18443, api :18080, grpc :50061, monitor :18090). Deploy via bash scripts/nezha-deploy.sh reboot ON nezha (rsync changed source to nezha FIRST). cmd/server (:18443/:8443) has a TRANSIENT startup health-window (HTTP/3 QUIC binds slower than the first health probes) — SIGQUIT dump proved it starts clean; wait ~60s then it's healthy.
- Build/vet/gofmt clean on the providers fix; full pkg/api suite GREEN; HTQ-FIX-006 guard registered.

NEXT (priority order)

1. **Independent §11.4.142 review of the bug-class fixes** (4fbe581 6-site Translate-arg data-loss + 7cabe3f MINOR-W6-1) if not already GO-reviewed in their own stream.
 2. **Full §11.4.40 retest from the last release tag to HEAD** (pre-build + post-build + nezha live re-validation of every fix's runtime signature + meta-test mutation sweep + Challenge bank incl. HTQ-FIX-006) → release-readiness.
 3. **Operator tag decision** (§11.4.151 prefix helix_translate-; §11.4.126 release-scope terminal condition; §11.4.113 FF-only no-force-push).
- Owed/tracked: re-record live /api/v1/providers for a fresh §11.4.153 Confirmed; markdown-translator EPUB→MD re-record (bug fixed at 4fbe581); continue §11.4.153 video coverage toward the ~remaining eligible features.
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 (prior) RESUME HERE (2026-06-16 ~15:00) —
 NEZHA HEAVY-TESTING ARC: api-400 fix +
gRPC round-trip + coverage + commentary-escape
fix (all live-validated)

 NEZHA HEAVY REAL-SERVICE TESTING — iterations 1→4
(live stack, sink-side, no bluff)

Heavy testing against the LIVE nezha stack (server-TLS :18443, api :18080, grpc :50061). Live image progression: f3904ccd (pre) → 2d0c925 (api-400 fix) → ebb82df (commentary-escape fix; server+grpc+api all on it now, HEALTHY).

- **iter1 5227585** — LLM commentary contamination (trailing meta-prose/ (Note:...)) + create-default-config env-override skip. FIXED + guarded.
- **iter2 2f326cd** — verifier dead-routes 404→honest 503; api /health IDLE false-negative (active gRPC probe); durable --env-file deploy (#3).
- **iter3 (salvage) 6a1aa8c** — api: reject unsupported provider with **400** instead of silent bridge-substitution (§11.4.69/§11.4.120). **RUNTIME RE-VALIDATION DONE (§11.4.108 PENDING→DONE)**: rebooted server on nezha (f3904ccd→2d0c925) and proved sink-side on POST https:// nezha.local:18443/api/v1/translate: RED pre-fix provider:"unsupported-provider"→HTTP **200** silent-sub to llm-novita; GREEN post-fix→HTTP **400** {"error":"unsupported provider: ..."}; VALID deepseek→200 real Serbian Добро јутро, како стаса?; default(empty)→200 Књига лежи на столу. (no regression). Evidence docs/ qa/ nezha_revalidate_6a1aa8c_20260616T142656Z/.
- **gRPC round-trip (this run)**: new cmd/grpc-translate-probe (standalone client) drove StartTranslation(/tmp/grpc_in.txt→/tmp/grpc_out.epub, ru→sr, deepseek)→poll GetTranslationStatus running→completed (3m21s)→read output. REAL Serbian Добар дан. Зовем се Иван. Живим у великом граду. ... + valid EPUB (mimetype/container.xml/content.opf/ toc.ncx/chapter1.xhtml) containing the translation. Evidence docs/ qa/ nezha_grpc_roundtrip_20260616T143649Z/.
- **Coverage (this run)**: 5 language pairs (ru↔sr, en→sr, ru→en) + script-conversion (sr latin Danas čitajem...) all 200 real; 24-parallel sustained load = 20×200 / 4×429 (clean rate-limit {"error":"Rate limit exceeded"}, p95~1.85s, 2s wall). Evidence docs/ qa/ nezha_coverage_20260616T144123Z/.
- **iter4 e198b17 (this run) — commentary-escape FIX (§11.4.4 test-interrupt → §11.4.102 → §11.4.115 → §11.4.135)**: the coverage batch surfaced 1/20 contamination the iter1 fix MISSED — Я пью кофе утром.→'Ја пијем кафу јутро.\n\n(Using Ekavica dialect and pure Serbian vocabulary as per guidelines)'. FACT root cause: isCommentaryBlock only stripped a trailing ()/[] block when inner started with note or contained translat; this style/dialect aside matched neither. FIX: widened with commentaryParenSignalWords (using/dialect/vocabulary/guideline/register/as per/i used/...); genuine in-content parentheticals preserved (new guard test). Anti-bluff: 2 live-captured RED fixtures FAIL pre-fix/PASS post-fix + KeepsBenignTrailingParenthetical over-strip guard + §1.1 mutation (remove signal-loop→FAIL→restore→PASS); pkg/translator/llm+pkg/ translator GREEN; vet clean. Deployed (server+grpc+api→ebb82df), re-validated live: 18/18 clean (the exact pre-fix-contaminating source now

CLEAN), task-1 400-fix unregressed. Evidence docs/qa/ nezha_revalidate_e198b17_20260616T145411Z/.

◆ OPERATOR REVIEW QUEUE (heavy-testing arc)

1. **.env.nezha line 48 malformed** (§11.4.10 — value NEVER printed/committed): python-dotenv could not parse statement starting at line 48 warns on every nezha deploy ⇒ OPENAI/ANTHROPIC keys may not be injected via that path (note: env NAMES OPENAI_API_KEY/ANTHROPIC_API_KEY ARE present <set> in containers, so injection is partial/via another mechanism). **Action:** operator reformats .env.nezha to one NAME=value per line (no continuation/quotes spanning lines around line 48).
2. **Live LLMsVerifier upstream not cross-network wired** ⇒ /api/v1/verified-models cannot serve real upstream data; bridge model-selection at server startup blocks ~3.5min on bridge.open (5m30s timeout) before binding (NOT a crash — it binds + HEALTHY). **Action:** enable the nezha LLMsVerifier upstream wiring + set its APIURL so /api/v1/verified-models returns real data and startup model-selection resolves fast.
3. **scripts/nezha-deploy.sh reboot dependency-ordering bug:** rebooting a service that has dependents (e.g. grpc-server, which api+server depend_on) fails podman rm ("has dependent containers") and then compose up collides on existing names ⇒ the dependent service silently keeps the OLD image (§11.4.108 risk — caught this run, recovered by stop+rm dependents-first then up). **Action:** make reboot stop+rm the dependent subtree before rm'ing a depended-on service (or use podman rm --depend/up --force-recreate once supported).

▶ (prior) RESUME HERE (2026-06-16 ~15:35) —
● NEZHA DISTRIBUTED BOOT COMPLETE
(full stack live + real translation verified) · §11.4.153
video program 43 confirmed · release 2.3.0 published

● NEZHA DISTRIBUTED BOOT — DONE + SINK-SIDE
VERIFIED (operator-CRITICAL task, COMPLETE)

The whole HelixTranslate System is DEPLOYED + RUNNING on nezha.local (Linux 6.12, 8c/62GB, podman 5.7.1, SSH milosvasic keyed) via the **containers submodule** (§11.4.76: launcher containers/cmd/deploy-stack commit 20173e8, parent gitlink 6296c03; Phase A defs 2166d2c; JWT fix 1b29668). **All 6 services Up HEALTHY:** helixtranslate-
{postgres,redis,monitor[:18090],grpc[:50061],api[:18080],server-TLS[:18443]}
(ports remapped to avoid host conflicts). **REAL translation round-trip VERIFIED sink-side (§11.4.69):** POST https://nezha.local:18443/api/v1/translate en→es → HTTP 200, provider llm-novita → "El viejo faro se alzaba solitario en la rocosa orilla." Secrets in gitignored .env.nezha on nezha (real JWT_SECRET + all provider keys, chmod-safe, NEVER committed §11.4.10). Stack left RUNNING for heavy testing. NEXT: run the heavy/real-production-service test suites against the live nezha endpoints; any bug → systematic-debugging → fix → rebuild → reboot → re-validate (NO bluff). ⚠ I premature-

stopped progressing agents TWICE during the boot — remote ops are 30-60+min;
BE PATIENT, trust completion, suspect hang ONLY if no completion + no ssh/
podman/curl subprocess + zero nezha state-change after 60+min.

§11.4.127 SHORT resumption (paste this first sentence into a fresh session)

Read docs/CONTINUATION.md (this file) + .remember/remember.md FIRST, then
git fetch --all --prune; **NEZHA BOOT COMPLETE** (full stack live on
nezha.local via containers submodule, real translation verified — see green block
above; run heavy testing against it next); the no-local-runtime release
helix_translate-2.3.0 is PUBLISHED (tag on main + 10 owned submodules;
helix_qa tag OWED §11.4.119); the **§11.4.153 per-feature video-confirmation
program** continues — docs/features/Status.md Rev 14, **43/496 features
video-confirmed** (waves 1+2+3a: bridge-ensemble/catalogue, DOCX/PDF/gRPC/
HTML-MD, providers novita/mistral/groq/siliconflow/cerebras/sambanova — real
bridge responses, ffprobe+content-verified, NO bluff), model source =
LLMsVerifier bridge (real keys present), HelixAgent incorporated (helix_agent/
pinned) NOT integrated (42-dep graph = deferred operator decision). **A release-
critical default-path defect the video program CAUGHT is FIXED + re-
reviewed GO + guarded: commit 7c2118a** (VerifiedFactory routes verifier-
selected models through the whitelist-immune path whenever a key exists from
ANY source; env-independent §11.4.138 guard; proven GREEN with all 36
*_API_KEY unset). **⚠ RECORDING ROOT-CAUSE (corrected via systematic-
debugging — do NOT repeat the earlier thrash):** recording is INTRINSICALLY
SLOW (~5–6 min/video: real LLM call + asciinema→agg→ffmpeg + ffprobe +
frame-analysis + agent reasoning). The bridge is NOT slow/hung (best-model=1s
cached; 600s provider timeout is intentional for long translations). Long stale-output
gaps are NORMAL — do NOT misread them as "hung" and premature-stop (that
churn killed progressing agents, e.g. wave-3a was stopped AFTER it had
committed). DISCIPLINE: dispatch small recording units (2–4 videos), be
PATIENT (give 25–30 min before even suspecting a hang), check the DECISIVE
signal (did it commit / are mp4s landing / is there an asciinema/ffmpeg/translate
subprocess) before stopping. Full autonomous coverage of ~90 remaining eligible
features ≈ many hours — a long steady iterative program, not a sprint. Continue the
loop (§11.4.126), no force-push (§11.4.113), no git add -A (§11.4.30), never touch
helix_qa (§11.4.119)/helix_agent build; SINGLE main-checkout committer at a time
(§11.4.84 — reviews read-only, recorders worktree-no-commit); contested checkout
(helix_code session shares machine) — verify branch/lock/sync before every
commit. NEXT: wave-3 owed (providers deepseek/cerebras/sambanova/hyperbolic/
gemini/hipu/cohere/fireworks/openrouter + REST endpoints + verify-models +
workable-items), then continue toward full coverage; web/desktop = honest
operator-attended (§11.4.3/§11.4.52). Tracked: thread CLI -max-tokens through
bridgeTranslator; commit untracked docs/design/
LLMSVERIFIER_BRIDGE.md/.html; clean stale worktree .claude/worktrees/agent-
a8ea4f6230cb2ee40.

§11.4.127 FULL resumption (live-state anchors — verified this session against git)

- **Parent HEAD 3ce413e** on main, local == origin/main (verified git fetch
--all this session; pushed FF to both upstreams github+HelixDevelopment).

- Untracked build/* + root binaries are EXTERNAL build-artifact churn (a gemini-cli/data-agent session rebuilds them) — NOT committed (§11.4.30).
- **⚠️ CONTESTED CHECKOUT (§11.4.84/§11.4.119):** an EXTERNAL gemini-cli/data-agent process intermittently runs `git checkouts` + has left a stale `.git/index.lock` in this repo. BEFORE EVERY commit: (a) confirm `git symbolic-ref -q HEAD == refs/heads/main`, (b) `ls .git/index.lock` absent (remove a STALE lock only if no live git process), (c) `git fetch --all + confirm local == origin/main`. Every R-5 commit survived via this protocol.
 - **R-5 commits this session (all FF-pushed):** 102c773 CM-NO-LOCAL-RUNTIME gate + paired meta-test · abc0912 Part-B NITs (bridge/format test asserts + bridgeOpener seam + detector XHTML anchor) · 7036a30 config.worker.json default_provider ollama→openai · 3ce413e §11.4.153 ledger 39 rows Obsolete + verifier-doc framing fix.
 - **Prior phase-2 commits (P-1..R-4):** 8325f9d R-4 remove worker configs + SSH/llama/ollama scripts · 86175d0/earlier R-4 remove SSH binaries + pkg/sshworker + pkg/modelsbridge + llamacpp_provider + markdown bridge wiring; P-1 BestClient e5307ce.

Bridge phase-2 — the CRITICAL no-local-runtime mandate (state: NEAR-COMPLETE)

Operator-CRITICAL: NO local runtime — only LLMsVerifier strongest models via pkg/bridge, bridged to components AND this Claude Code session. Plan: **docs/design/BRIDGE_PHASE2_EXECUTION_PLAN.md** (+ design docs/design/LLMSVERIFIER_BRIDGE.md, removal map docs/design/LOCAL_RUNTIME_REMOVAL.md).

LOCKED operator decisions: D1 remove local runtimes entirely; D2 forbid ALL local runtimes (llama.cpp + Ollama); R1 → **R-4 decision (2026-06-16): keep the distributed/API path, remove SSH-local** (the SSH-local binaries + worker body Obsolete §11.4.90; version_manager/ssh_pool/distributed coordination KEPT); R2 **require API keys everywhere** (no offline/local fallback — honest error/SKIP, never a silent local fallback); D4/D5 **top-1 + fallback, in-process; ensemble = "Provider-diverse verified"**.

Status — DONE:

- **P-1 / R-1** BestClient/BestTranslator + all default-path components redirected to the bridge.
- **R-2** Ollama removed · **R-3** llama.cpp removed · **R-4** SSH-local binaries + pkg/sshworker + pkg/modelsbridge + local worker configs/scripts removed (operator-confirmed).
- **R-5 (this session):** CM-NO-LOCAL-RUNTIME pre-build gate (3 arms, default-path-only) + paired §1.1 meta_test_no_local_runtime.sh (baseline PASS / Mut1 Arm1 FAIL / Mut2 Arm3 FAIL / Neg PASS); Part-B NITs; config.worker.json fixed; §11.4.153 ledger 39 rows Obsolete (docs_chain features re-synced HTML/PDF/DOCX); AGENTS.md + README.md + this CONTINUATION updated.

- **§11.4.40 FULL RETEST: PASS** (authoritative main-checkout, HEAD f9c52a2; evidence docs/qa/bridge_phase2_retest_20260616/SUMMARY.md, commit d1bed61): sweep all-9-gates green · build clean · functional 55 pkgs OK · -race only pre-existing test/distributed (§11.4.7) · meta-tests all bite (incl. meta_test_no_local_runtime.sh) · determinism stable · product challenges pass. Two prior worktree-retest FAILs proven ENVIRONMENT artifacts (worktree missing submodules; worktree checkout mtime) — same gate PASSES on the real main checkout; NOT regressions.

PHASE-2 CLOSED + RELEASED. 🟠 **helix_translate-2.3.0** (operator "do tag now" 2026-06-16; safest zero-churn calls: version = current VERSION 2.3.0 no bump; set = main + 10 owned submodules, helix_qa excluded §11.4.119). Annotated tag created + pushed to ALL remotes across **main (6b81a10) + containers/challenges/doc_processor/llm_orchestrator/llm_provider/vision_engine/llms_verifier/constitution/docs_chain/security** (each at its gitlink-pinned commit), verified via ls-remote (main ^{}→6b81a10). **OWED:** helix_qa tag (off-limits to this session §11.4.119 — operator/its session to apply helix_translate-2.3.0).

Non-blocking follow-ups (not product defects): (a) host_no_auto_suspend challenge FAILs = HOST not hardened against auto-suspend — environment (§11.4.3), host power config NOT changed autonomously (CONST-033), operator action if desired; (b) internal/working/config.distributed.{json,test,thinker}.json legacy llamacpp/gguf tokens (KEPT distributed path) — future R-cycle if distributed worker API-migrated; (c) codebase_hash_report.json generator needs a submodule-exclusion before faithful refresh (not a gate dep); (d) format-detector XHTML-branch advisory (conservative, zero real-tree impact) — optional future hardening.

Standing constraints (binding every session)

- **No force-push EVER** (§11.4.113 — merge-onto-latest-main, FF-only); **never git add -A** (§11.4.30); **never touch the helix_qa submodule** — another session owns it (§11.4.119 single-resource-owner); **never touch owned submodules** in this R-5 scope.
 - **No host power-management** (§12 / CONST-033). **Require API keys, NO local llama.cpp/Ollama/SSH-distributed runtimes.** Operator-blocked → AskUserQuestion (§11.4.66). Endless autonomous loop is the DEFAULT (§11.4.126).
 - **Working API keys:** DEEPSEEK_API_KEY + GROQ_API_KEY present (real translations work); GEMINI/ZHIPU invalid/quota-blocked (honest errors). NEVER print key values (§11.4.10).
 - **Contested checkout:** apply the pre-commit vigilance protocol above before every commit (§11.4.84/§11.4.119).
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🚫 (SUPERSEDED 2026-06-15 ~22:30 — subagent runtime has since reset; reconciliation + P-1 landed) RATE-LIMIT BLOCK + OWED CRASHED- STREAM WORK (2026-06-15 ~19:05)

The subagent runtime hit a session/usage limit (subagents die instantly, ~178 tokens). NO new subagents can run until the **7:10pm Europe/Moscow reset**. Two phase-1 bridge streams CRASHED mid-work (§11.4.147 crash≠done — work OWED, partial state PRESERVED uncommitted in the worktree, do NOT discard, do NOT commit-incomplete §11.4.84/§11.4.121):

LLMsVerifier-bridge mandate (operator CRITICAL 2026-06-15): no local llama.cpp — only LLMsVerifier strongest models + bridge to components AND this Claude Code session. Design: docs/design/LLMSVERIFIER_BRIDGE.md; removal map: docs/design/LOCAL_RUNTIME_REMOVAL.md. LOCKED operator decisions: D1 remove llama.cpp entirely; D2 forbid ALL local runtimes (llama.cpp + Ollama + SSH/distributed); D3 agent access = CLI + MCP; D4/D5 top-1 + fallback, in-process pipeline (env *_API_KEY keys, no service); R1 Obsolete cmd/translate-ssh + cmd/ssh-translation + llamacpp/ollama worker configs (§11.4.90); R2 require keys everywhere (no offline path — tests FAIL not skip when keys absent).

OWED CRASHED STREAMS (respawn after reset, §11.4.147 + §11.4.84 quiescence-check first):

- **bridge-core** (was a9659f5) — INCOMPLETE, does NOT build: cmd/model-bridge/main.go:52: undefined: runMCP (crashed before writing the MCP stdio server). Preserved partial: pkg/bridge/ (BestTranslator/Invoke/BestModel/ListVerified facade), cmd/model-bridge/main.go (CLI), internal/verifier/provider_resolver.go+_test.go (numeric-ProviderID gap-fix). RESPAWN to: finish runMCP (MCP stdio server) + .mcp.json wiring + full tests + real best-model/invoke evidence + docs/guides/LLMSVERIFIER_BRIDGE_USAGE.md; then §11.4.142 review + commit.
- **web-UI** (was aaf872a) — BUILDS clean (pkg/api + cmd/server). Preserved: pkg/api/dashboard.go+dashboard_test.go, pkg/api/handler.go (M) — wires the unrouted Web Dashboard translation UI (the real §11.4.153 web gap). OWED: independent §11.4.142 review + real-server real-translation evidence run, then commit. (Note: a later phase swaps its provider to bridge.BestTranslator.)

NEXT (post-reset, phased per LOCAL_RUNTIME_REMOVAL.md R-0→R-5): (1) respawn bridge-core to finish; (2) review+commit web-UI; (3) phase-2: redirect all 9 components to bridge.BestTranslator, remove Ollama, remove llama.cpp, Obsolete SSH/distributed (R1), add CM-N0-LOCAL-RUNTIME gate + paired mutation, require-keys-everywhere (R2); (4) HelixQA web-runner browser+video backend extension (needs the helix_qa session) → then the real web-dashboard video. Web-dashboard video remains PENDING until both the UI wiring lands AND HelixQA gains a web/video backend.

DONE + PUSHED this session (all remotes, FF-only): constitution §11.4.153 (74e7fbe, 10 remotes); parent 58178a5→a5e8866→86bf81e→d53e085→fb74c2d→87cd2be→e73e0b7→fb265e7; submodules llm_provider 724e284, containers 2a9923a, llms_verifier 89719e2b.

§11.4.153 mandate + 470/472 ledger (21 video-confirmed) + DOCX & PDF output writers + multipass wiring + cli title/atomic-write fixes + prep-translator/server/REST bug fixes + README §11.4.57 + all 9 submodules GREEN.

✓ §11.4.153 LEDGER RE-SYNC (2026-06-15, +DOCX output + -multipass video + cli atomic-write)

Three capabilities became real+evidenced this session and landed+pushed; the feature ledger (docs/features/Status.md + Status_Summary.md) is re-synced honestly (docs UNCOMMITTED — conductor commits, no git add -A):

- **DOCX output** — gap → **Implemented** (pure-Go OOXML writer pkg/ebook/docx_writer.go, commit 87cd2be). Verified this session by re-producing a .docx via the shipped writer → file = Microsoft Word 2007+, all 4 OOXML parts, real translated content, correct XML escaping. **Real-artifact-confirmed (NOT video — no .mp4 yet, video PENDING).**
- **-multipass multi-pass verification** — now **video-confirmed**: helixtranslate-cli-multipass-verify-20260615.mp4 (ffprobe 888x630, 93 frames, 9.3s) shows "Step 4: Multi-pass Polishing ✓ Success — Polished over 1 pass with deepseek" + real Spanish output. Engine wired into unified-translator via -multipass in d53e085.
- **cmd/cli atomic-write / no-clobber** (87cd2be) — write-safety fix, guard cmd/cli/no_partial_output_test.go (unit-proven, no video required); title path-leak fixed in d53e085.
- **Honest counts**: video-confirmed **20** → **21** (the -multipass flip) + **1 real-artifact-confirmed** (DOCX); Status/Status_Summary Rev 2→3; not-implemented gap rows 2→1.
- **Exports re-synced**: docs_chain sync features (6 derived nodes) → docs_chain verify features = **in-sync (RC=0)**; HTML+PDF+DOCX siblings regenerated for both docs (all file-verified).
- **Remaining gaps now**: PDF output (no writer); cmd/translator local path STUB; llm_provider gofmt hygiene (23 flagged files); cosmetic docx title-twice; web-dashboard translation video **OPERATOR-BLOCKED** (HelixQA-gated). translate-ssh/ebook-translator stay OPERATOR-BLOCKED.

✓ GAP-CLOSURE BATCH (2026-06-15, video/coverage-surfaced gaps — all reviewed GO + pushed)

Three gap-closures (each RED-first §11.4.115, mutation-verified, independent batch review VERDICT GO §11.4.142) landed + pushed:

- **Parent d53e085** (pushed all remotes): (1) **cmd/cli title path-leak FIXED** — TXT parser set Metadata.Title=filepath → translated → leaked into output EPUB <dc:title>; sanitizeBookTitle() added at the cmd/cli input→book seam (genuine titles pass through; real run <dc:title>Tiha bašta</dc:title>, NO LEAK). (2) **pkg/verification multipass engine WIRED** (§11.4.124 — git-history proved it was NEVER imported by any cmd/, born dormant 717bf18, finished real code not scaffolding) via opt-in -multipass

flag (default OFF, back-compat; guardrails: passes $\geq$ 1, base preserved on error/empty); RED→GREEN httptest + real DeepSeek "Step 4: Multi-pass Polishing "; §11.4.150 research docs/research/20260615_video_surfaced_fixes/verification_multipass_wiring.md.

- **llm_provider 724e284** (pushed LLMPProvider remote; parent gitlink bumped in d53e085): contract test for the interface-only pkg/provider (zero-coverage §11.4.28 gap; mutation-proven; suite green).

NEXT (open gaps, tracked — none blocking the current scope): implement DOCX/PDF **output** writers in pkg/ebook (input-only today); implement cmd/translator local path (STUB "not yet implemented"); fix cmd/cli silent-fail-exit-0 when -model omitted (incidental finding); record a -multipass video for §11.4.153 ledger confirmation (currently real-run+RED→GREEN evidenced, no .mp4 yet); gofmt the 23 pre-existing flagged files in llm_provider; web-dashboard translation video is **OPERATOR-BLOCKED** (HelixQA-gated, separate session). translate-ssh/ebook-translator stay OPERATOR-BLOCKED (need remote SSH/llama.cpp host).

§11.4.153 FEATURE-STATUS VIDEO-CONFIRMATION SYNC (2026-06-15, ~25 real videos → ledger honest)

The §11.4.153 per-feature Status ledger is now in honest sync with the real video evidence captured this session (docs uncommitted — conductor to commit; do NOT git add -A).

What landed before this sync (already pushed):

- **§11.4.153 doc set** (docs/features/Status.md + Status_Summary.md, 470/472 features, four-format HTML+PDF+DOCX) shipped in 58178a5; constitution bump 74e7fbe.
- **3 video-surfaced bugs FIXED + pushed** in a5e8866 (the green suite had missed all 3, §11.4.4/§11.4.138): (1) preparation-translator DEAD — DeepSeek API key never reached config (-api-key/env not wired) → "DeepSeek API key is required", no analysis JSON; (2) cmd/server TLS startup — config with no tls_cert_file failed to start; fix backfills self-signed certs in LoadConfig; (3) REST POST /api/v1/translate ignored target_lang (hardcoded) → always Serbian regardless of requested language.
- **~25 real videos** captured under /Volumes/T7/Downloads/Recordings/helixtranslate-*.mp4 (all ffprobe-verified dur>0+frames>0; key ones content-verified frame-by-frame this session).
- **All 9 owned submodules GREEN** — coverage logs qa-results/submodule_coverage_20260615/.
- **README §11.4.57** Tracked-Items doc-link section added (prior session).

This sync (Status.md/Status_Summary.md Rev 1→2 + exports re-synced):

- **Video-confirmed: 1 → 20** feature rows ($\approx$ 4.3% of 472), each flipped to Confirmed: ONLY after the cited .mp4 was content-verified (§11.4.6 no overclaim). Confirmed: unified-translator (DeepSeek translate; EPUB→TXT / HTML→EPUB / FB2→EPUB / FB2→FB2; Serbian-Cyrillic; -script Cyrillic[?]Latin; format-detection; help/version; -verify), REST API (POST /

api/v1/translate real DeepSeek + target_lang FIXED, TLS startup FIXED, /health/version/providers), gRPC (real EN→ES), markdown-translator (EPUB→MD round-trip), preparation-translator (FIXED), cmd/cli (real DeepSeek), monitor-server (live WS hub), DeepSeek library client.

- **Honest gaps recorded (§11.4.6, NOT confirmed):** DOCX output + PDF output UNIMPLEMENTED (no writer in pkg/ebook; unified-translator rejects them; 2 new gap rows added → 472 total); -verify ≠ pkg/verification multipass engine (engine has no CLI entry point wired in); cmd/translator local path = STUB ("local translation not yet implemented"); cmd/translate-ssh + cmd/ebook-translator = OPERATOR-BLOCKED (need remote SSH/llama.cpp worker host); GET /api/v1/verified-models = 404 (LLMsVerifier disabled in config); cmd/cli minor cosmetic file-path-leak into title metadata.
- **Exports + fingerprint re-synced** via docs_chain sync features (applied: 6 derived nodes — HTML/PDF/DOCX × 2 docs) → docs_chain verify features = **in-sync** (RC=0); evidence qa-results/docs_chain/20260615T144104Z/sync.log; docs_chain state.json = §11.4.86 content-hash fingerprint store.

NEXT (open gaps queued): implement DOCX/PDF output writers in pkg/ebook; wire pkg/verification multipass engine into a CLI entry point; implement cmd/translator local path; produce videos for remaining PENDING runtime features (web dashboard translation, other providers) to lift coverage. translate-ssh/ebook-translator stay OPERATOR-BLOCKED until a remote SSH/llama.cpp host is provided.



WAVE 15 — SECOND-PASS + determinism (campaign 101→110; second pass STILL productive)

Parent HEAD 6c89910, build/vet/test green, git OK, llms_verifier pointer intact.

- **pkg/ebook 6c11dee (#102):** EPUB parser set chapter.Title to the zip entry name → EPUB→translate→EPUB rendered "OEBPS/chapter1.xhtml" as every chapter heading + TOC entry (core use case). Fix = extractChapterTitle (